

Part I

SUMMARY of Linear Approximation and Differentials

Summary

February 22, 2026

Definition

If a function f is **differentiable** at $x = a$, then a **linear approximation to f at a** is a function given by

$$L_a(x) = f(a) + f'(a)(x - a)$$

Note 1: When the value of a is understood, the subscript is sometimes dropped from the notation. In this case, it is written as just $L(x)$ instead of $L_a(x)$.

Note 2: The graph of L is the line tangent to the graph of f at the point where $x = a$.

Note 3: If x is near a , the value $f(x)$ can be approximated by the value of $L(x)$.

Note 4: If the graph of f is concave down *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an overestimate. If the graph of f is concave up *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an underestimate. (The concavity has to be consistent across the interval, not just at a point.)

A **differential**, df of f at x is given by

$$df = f'(x)dx$$

Note 3: If we consider a point x and if L is the linear approximation of f at x , then for any point $x + dx$ the following holds

$$f(x + dx) \approx L(x + dx) = f(x) + f'(x)dx = f(x) + df$$

Note 4: The **increment** of f , Δf , is given by

$$\begin{aligned}\Delta f &= f(x + dx) - f(x) \approx L(x + dx) - f(x) = \\ &= f(x) + f'(x)dx - f(x) = f'(x)dx = df\end{aligned}$$

Therefore, $\Delta f \approx df$.

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 (a) Find the linearization, $L(x)$, of the function $f(x) = e^{2x}$ at $a = 0$.

Explanation. Recall that $L(x) = f(a) + f'(a)(x - a)$.
Hence $f'(x) = 2e^{2x} \implies f'(0) = 2e^0 = 2$ and $f(0) = e^0 = 1$.
Therefore $L(x) = 1 + 2 \cdot x$

(b) Using the linearization, $L(x)$, from the part (a), approximate e .

Explanation.

$$\begin{aligned} e &= e^{2 \cdot (1/2)} \\ &= f(1/2) \approx L(1/2) \\ &\approx 1 + 2 \cdot (1/2) \\ &\approx 2 \end{aligned}$$

(c) Is the estimation found in part (b) an overestimate or an underestimate?
EXPLAIN.

Explanation. $f''(x) = 4e^{2x}$ is positive on the interval $\left(0, \frac{1}{2}\right)$. That means the graph of f is concave up between $x = 0$ and $x = \frac{1}{2}$, so this estimate is an underestimate. (The graph of the linearization L lies below the graph of the function f .)

Problem 2

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Problem 2 Complete steps (i)-(vii) below in order to estimate the following values using linear approximation:

(a) $\cos\left(\frac{31\pi}{180}\right)$

(b) $\sqrt[3]{8.13}$

- (i) Identify the function, $f(x)$.
- (ii) Find the nearby value where the function can be easily calculated, $x = a$.
- (iii) Find $\Delta x = dx$.
- (iv) Find the linear approximation, $L(x)$.
- (v) Compute the approximate value of the expression using the linear approximation.
- (vi) Compare the approximated value to the value given by your calculator.
- (vii) Compare dy and Δy using the value given by your calculator.

Explanation. (a) $\cos\left(\frac{31\pi}{180}\right)$

- (i) $f(x) = \cos x$
- (ii) $a = \frac{30\pi}{180} = \frac{\pi}{6}$
- (iii) $\Delta x = \frac{31\pi}{180} - \frac{\pi}{6} = \frac{\pi}{180}$
- (iv)

$$\begin{aligned} L(x) &= f\left(\frac{\pi}{6}\right) + f'\left(\frac{\pi}{6}\right)\left(x - \frac{\pi}{6}\right) \\ &= \cos\left(\frac{\pi}{6}\right) - \sin\left(\frac{\pi}{6}\right)\left(x - \frac{\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(x - \frac{\pi}{6}\right) \end{aligned}$$

(v)

$$\begin{aligned} L\left(\frac{31\pi}{180}\right) &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{31\pi}{180} - \frac{\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{\pi}{180}\right) \\ &= \frac{1}{2}\left(\sqrt{3} - \frac{\pi}{180}\right) \\ &\approx 0.857299 \end{aligned}$$

$$(vi) \cos\left(\frac{31\pi}{180}\right) \approx 0.857167$$

(vii)

$$dy = L\left(\frac{31\pi}{180}\right) - L\left(\frac{\pi}{6}\right) \approx -0.008727$$

$$\Delta y = \cos\left(\frac{31\pi}{180}\right) - \cos\left(\frac{\pi}{6}\right) \approx -0.008858$$

$$(b) \sqrt[3]{8.13}$$

$$(i) f(x) = \sqrt[3]{x}.$$

$$(ii) a = 8.$$

$$(iii) \Delta x = 8.13 - 8 = 0.13.$$

(iv)

$$\begin{aligned} L(x) &= f(8) + f'(8)(x - 8) \\ &= \sqrt[3]{8} + \frac{1}{3(\sqrt[3]{8})^2} (x - 8) \\ &= 2 + \frac{1}{12}(x - 8) \end{aligned}$$

(v)

$$\begin{aligned} L(8.13) &= 2 + \frac{1}{12}(8.13 - 8) \\ &= 2 + \left(\frac{1}{12}\right)\left(\frac{13}{100}\right) \\ &= 2 + \frac{13}{1200} = \frac{2413}{1200} \\ &\approx 2.010833 \end{aligned}$$

$$(vi) \sqrt[3]{8.13} \approx 2.010775.$$

(vii)

$$dy = L(8.13) - L(8) \approx 0.010833$$

$$\Delta y = \sqrt[3]{8.13} - \sqrt[3]{8} \approx 0.010775$$

Problem 3

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Problem 3 Estimate the value of $\sin\left(\frac{178\pi}{180}\right)$. Indicate whether your value is an overestimate or an underestimate.

Explanation. Set $f(x) = \sin(x)$. The problem is asking us to use linear approximation to estimate the value of $f\left(\frac{178\pi}{180}\right)$. Since $\frac{178\pi}{180}$ is very close to π , we will use L as the linearization of $\sin(x)$ at $a = \pi$.

$f'(x) = \cos(x)$ so $f'(a) = \cos(\pi) = -1$. $f(a) = \sin(\pi) = 0$. The linear approximation is given by $L(x) = -(x - \pi) + 0$.

$$\begin{aligned}\sin\left(\frac{178\pi}{180}\right) &\approx L\left(\frac{178\pi}{180}\right) \\ &= -\left(\frac{178\pi}{180} - \pi\right) \\ &= \frac{\pi}{90}.\end{aligned}$$

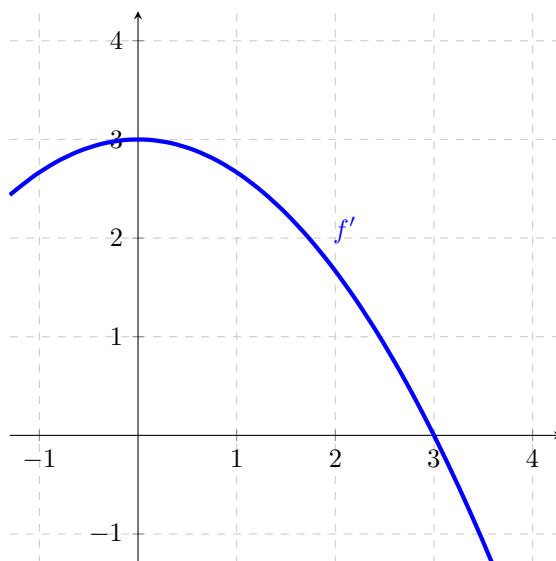
$f''(x) = -\sin(x)$. That means $f''(\pi) = 0$, BUT f'' is negative for all x in the interval $\left(\frac{178\pi}{180}, \pi\right)$. The graph of f is concave down across this interval, so the estimate is an overestimate.

Problem 4

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Problem 4

Problem 4 Consider the graph of $f'(x)$ given below. Suppose you know that $f(3) = 7$. Can you approximate $f(2.98)$ and $f(3.02)$? Explain your answer. Are these overestimates or underestimates?



Explanation. $f(3) = 7$ and $f'(3) = 0$

$$L(x) = 7 + 0(x - 3) = 7$$

This is a constant function, and so our approximations are $f(2.98) \approx 7$ and $f(3.02) \approx 7$. These are overestimates for the graph of f is concave DOWN on the intervals $(2.98, 3)$ and $(3, 3.02)$.

Problem 5

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Problem 5 Consider a square with a side x . Let A be the area of the square.

- (a) Compute ΔA , the change in area if the side increases by $\Delta x = dx$.

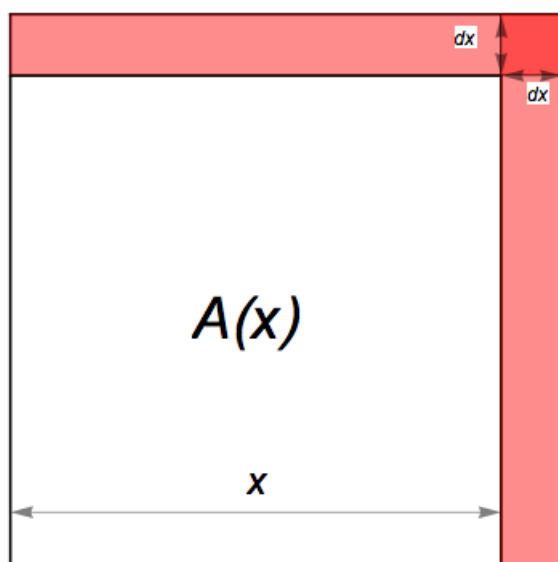
Explanation. $\Delta A = A(x + dx) - A(x) = (x + dx)^2 - x^2 = x^2 + 2x \cdot dx + (dx)^2 - x^2 = 2x \cdot dx + (dx)^2$

- (b) Compute dA , the differential of A at x , and compare it to ΔA .

Explanation. $dA = A'(x) dx = 2x \cdot dx$

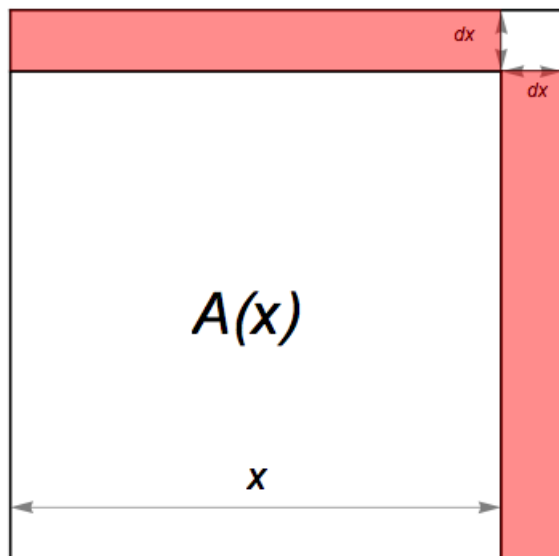
$\Delta A = [2x \cdot dx] + (dx)^2 = dA + (dx)^2$

- (c) In the figure below the shaded part represents the change ΔA . Shade the part that represents dA .



Explanation. Since $dA = A'(x) dx = 2x \cdot dx$, we shade the part as shown in the figure below.

Problem 5



The picture illustrates that $\Delta A \approx dA$.

Problem 6

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Problem 6 Estimate the amount of paint needed to apply a coat of paint .05 cm thick to a hemispherical dome with diameter 50m. Is this value an underestimate or an overestimate?

Explanation. The radius of the dome is $\frac{50}{2}m = 25m$. The paint increases this by 0.0005m (0.05cm to meters). The volume of a “hemispherical dome” (or half of a sphere) is $V = \frac{1}{2} \left(\frac{4}{3} \pi r^3 \right) = \frac{2}{3} \pi r^3$. Then

$$dV = 2\pi r^2 dr$$

The amount of paint needed is approximately the change in volume (dV). We also have that $r = 25m$ and $dr = (5 \times 10^{-4})m$. Thus, the amount of paint needed to paint the dome is approximately

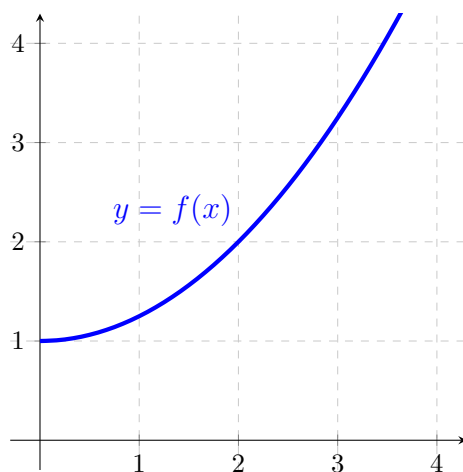
$$\left[dV \right]_{r=25, dr=0.0005} = 2\pi(25)^2(5 \times 10^{-4}) = 5\pi(0.125) = 0.625\pi \approx 1.9635m^3$$

The function $V(r) = \frac{2}{3}\pi r^3$ has derivatives $\frac{dV}{dr} = 2\pi r^2$, and $\frac{d^2V}{dr^2} = 4\pi r$. This second derivative is positive for r in the interval $(25, 25.0005)$, so the graph of V is concave up in that entire interval. This means the estimate is an underestimate.

Problem 7

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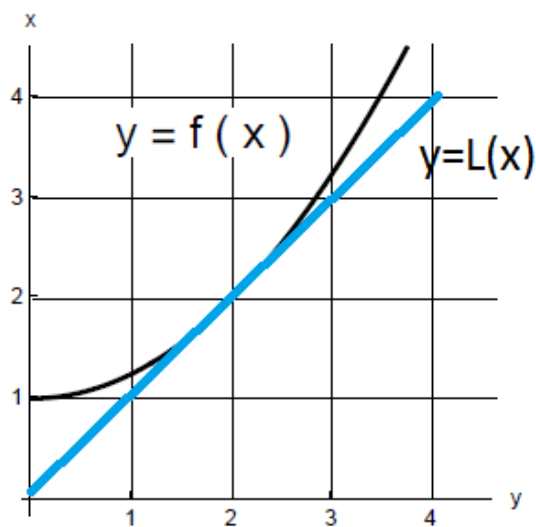
Problem 7 The graph of a function f is given below.



- (a) Given that $f'(2) = 1$, find the linear approximation L to the function f at $a = 2$. [2]

Explanation. $L(x) = f(2) + f'(2)(x - 2) = 2 + (x - 2) = x$

- (b) Sketch the graph of L in the figure above.

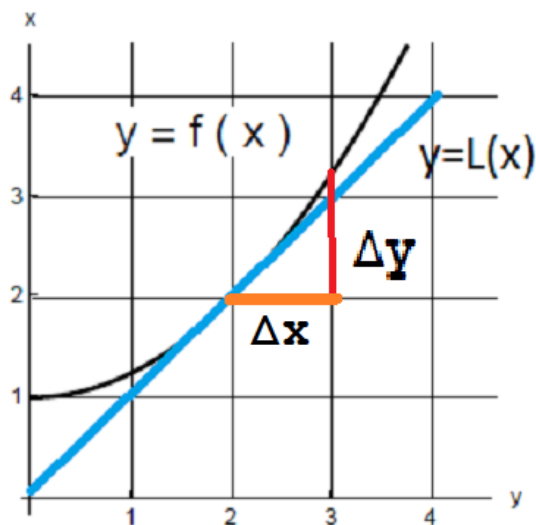


Explanation.

- (c) Use the linear approximation L to estimate the value of $f(3)$. Is this an underestimate or overestimate? **EXPLAIN.** [2]

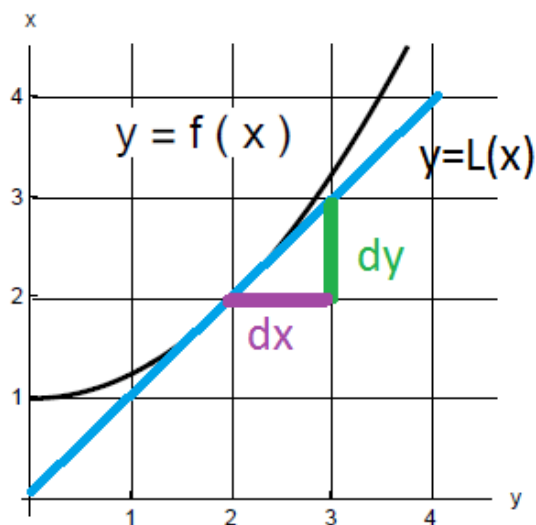
Explanation. $f(3) \approx L(3) = 3$. It is an underestimate because f is concave up on the interval $(2, 3)$.

- (d) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the **function** $y = f(x)$, Δy , is given by $\Delta y = f(a + \Delta x) - f(a)$. Draw and label Δy and Δx in the figure above.



Explanation.

- (e) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the **linear approximation**, dy , is given by $dy = L(a + \Delta x) - L(a) = f'(a)\Delta x$. Draw and label $L(x)$, dx and dy (differential) in the figure above.



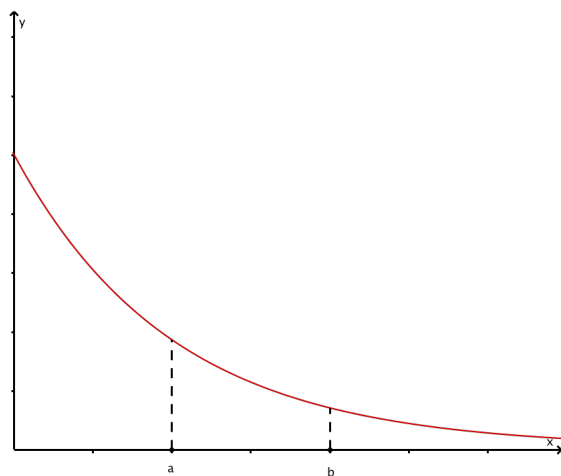
Explanation.

Problem 8

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Problem 8

Problem 8 The figure shows the graph of a function f . Let $L_a(x)$ be the linear approximation of f at a .

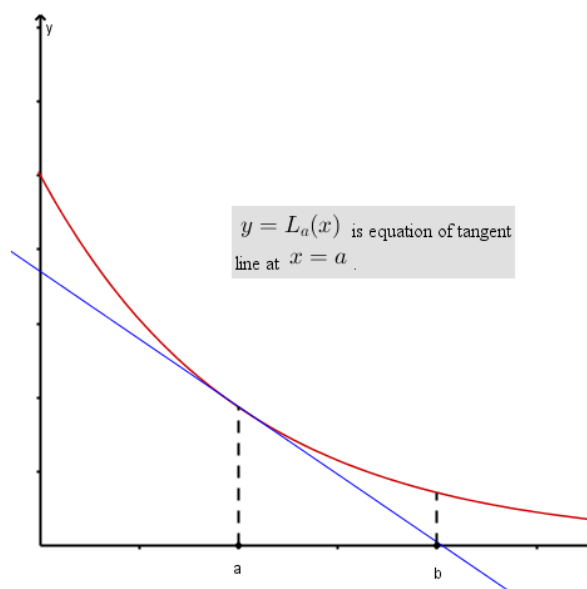


Circle ALL the correct statements below.

- (a) $L_a(b) < f(b)$
- (b) $L_a(b) > f(b)$
- (c) $L_a(a) < f(a)$
- (d) $L_a(a) > f(a)$
- (e) No statement (a) – (d) is correct.

Explanation. From the graph of f and the the graph of $L_a(x)$ we see that the correct statement is (a):

Problem 8



Problem 9

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Problem 9

Problem 9 By using linear approximation, determine which of the following is the best estimate of $e^{0.002}$.

(a) 1.00100050016679834166

(b) 1.00200200133400026675

(c) 1.00300450450337702601

(d) 1.02020134002675581016

Explanation. Let $f(x) = e^x$ and $a = 0$. Then since $f'(x) = e^x$, we have that

$$\begin{aligned} L(x) &= f(a) + f'(a)(x - a) \\ &= f(0) + f'(0)(x - 0) \\ &= e^0 + e^0 x \\ &= 1 + x \end{aligned}$$

Then since $L(0.002) = 1 + 0.002 = 1.002$, the answer is (b).

Problem 10

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Problem 10 Find a formula for the differential of the following functions.

(a) $y = 3x^6 e^x$.

Explanation. $\frac{dy}{dx} = 18x^5 e^x + 3x^6 e^x$ so

$$\begin{aligned} dy &= \frac{dy}{dx} dx \\ &= (18x^5 e^x + 3x^6 e^x) dx. \end{aligned}$$

(b) $z = \ln(1 + t^2)$.

Explanation. $dz = \frac{2t}{1 + t^2} dt$

(c) $\theta = \tan^{-1}(r^3)$.

Explanation. $d\theta = \frac{3r^2}{1 + r^6} dr$

Problem 11

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Problem 11

Problem 11 *In your own words, explain why $L_a(x)$ is a good approximation of the function f for x values $x \approx a$.*

Explanation. The graph of $L_a(x)$ is the line tangent to the graph of f at $(a, f(a))$. For a differentiable function, the tangent line is very close to the graph of the function for x -values near that point.